Prabhjot Singh

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EDUCATION

University of Ottawa, Master of Computer Science (Applied AI concentration) Sep 2026 - May 2028 (expected)

Delhi Technological University, BTech, Computer Software Engineering, CGPA 8.1/10 Jun 2022 - May 2026

Relevant coursework: Machine Learning, Methods of Data Analysis, Data Structures and Algorithms, DBMS, Operating Systems

EXPERIENCE

Infosys Springboard, AI Intern (Virtual Internship Program) Remote | Nov 2025 - Jan 2026

- Built an LLM-based legal advice assistant on web-scraped legal sources that I cleaned and organized for retrieval.
- Scored sources for reliability (1.0 government, 0.5 established outlets, 0.2 blogs/forums, 0 unknown) to favor trusted ones.
- Improved answer accuracy by about 23% by refining prompts and retrieval, which reduced hallucinations in legal answers.

VaultOfCodes, AI and Prompt Engineering Intern Remote | Jul 2025 - Aug 2025

- Built FastAPI endpoints and added sentence-transformer embeddings for semantic search in LLM-based features.
- Wrote a Python pipeline that extracts and processes text from PDF documents automatically.

AltQube, Data Science Intern Bengaluru, IN | Jun 2025 - Aug 2025

- Built customer segmentation models with K-Means and DBSCAN on customer portfolio data.
- Improved cluster quality by about 12% (Silhouette Score) by switching from a K-Means baseline to DBSCAN.
- Created features from past portfolio data to describe how customers invest.

PROJECTS

ARA: Claim-Level Verification for RAG *(code for my published paper)* LangGraph, BAAI/bge-m3, FAISS, Exa, Chainlit

- Built a LangGraph pipeline that splits an LLM answer into separate claims, searches the web for evidence on each one, and labels it SUPPORTED, INSUFFICIENT or CONTRADICTED.
- Stored the evidence as BAAI/bge-m3 embeddings in FAISS and ranked sources by how reliable the website is.
- Cut false "supported" labels by about 80% compared to a standard RAG setup (289 down to 56) on 496 hand-labeled claims from scientific papers.
- The same approach works for fact-checking news claims, since it will not confirm a claim it cannot find evidence for.

MeowTutor AI *(team project, I built the backend)* LangChain, Pinecone, Tesseract OCR

- Built the backend: Tesseract OCR for notes, LangChain multi-query search over Pinecone, and LLM-generated MCQs.
- Piloted with 100 university students.

Adobe India Hackathon 2025, Round 1 (1A, 1B) PyMuPDF, FAISS, TinyLlama, Docker

- 1A: Pulled titles and H1, H2, H3 headings out of PDFs using font sizes in PyMuPDF and saved them as JSON.
- 1B: Built an offline Docker app that summarizes the PDF sections most relevant to a user's role, using bge-small, FAISS and TinyLlama.

PUBLICATION

Singh, P. (2026). "A Claim-Centric Multi-Source Verification Architecture for Reducing Hallucinations in Retrieval-Augmented Generation." *Amity Journal of Computational Sciences*, Vol. 10, Issue 1. Read the paper

TECHNICAL SKILLS

Languages: Python, C++, C

ML / LLM: PyTorch, Hugging Face Transformers, scikit-learn, LangChain, LangGraph, sentence-transformers, FAISS, Pinecone

Data: pandas, NumPy, SciPy, Matplotlib, seaborn, Tesseract OCR

Backend & Tools: FastAPI, Flask, SQLAlchemy, MySQL, SQLite, Docker, Git, Jupyter, Google Colab

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Oracle Cloud Infrastructure 2025 Certified Generative AI Professional, Data Science Professional, and AI Foundations Associate; Google Cloud Gen AI Academy 2.0 (AI/ML Track); Amazon ML Summer School

Achievements: 97.85 percentile in the Naukri Campus Young Turks 2025 national skill contest; Top 75 coders, Amazon Hackathon

Languages: English, Hindi, Punjabi (fluent); French (beginner)

VOLUNTEER & LEADERSHIP

Volunteer Mentor, Desh Ke Mentor Program, Government of NCT of Delhi Jan 2023 - Apr 2023

- Mentored high school students one on one about choosing their academic stream and planning their careers.